

II B.E. (Information Technology)
Class Test I September 2022
Subject: ITR3C3 : Data Structures

Time: 70 min

Max Marks: 20

Note : Attempt all questions. Make and state suitable assumptions when it is required while answering. Write code with proper indentation and clarity.

- Q1 Write a function with following signature
 void rotate(node *t)

5

The function takes node pointer of "start" (first node of link list) and rotate the elements in the list as shown in following example:

List t { a1, a2, a3, a4, a5, a6 }

List after rotate { a2, a3, a4, a5, a6, a1 }

- Q2 Write a function with following signature
 node* alternateMerge(node *t1, node* t2)

5

The function takes two node pointers of "start" (first node of link list) and returns "start" pointer of resulting link list. Resulting link list is to be created using element values from the two lists (t1 and t2) in alternate fashion. Example :

List t1 { a1, a2, a3, a4 }

List t2 { b1, b2, b3, b4, b5, b6 }

Result List { a1, b1, a2, b2, a3, b3, a4, b4, b5, b6 }

PTO